TruthLens

Fake News Detection

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Motivation & SDG Mapping

Problem Motivation

- Social media platforms (Facebook, Twitter, WhatsApp) have become primary news sources for billions of people
- Fake news spreads 6 times faster and reaches more people than real news
- Manual fact-checking is too slow to keep up with the volume of online content

Real-World Importance

- Political manipulation through fake news during elections
- Health misinformation (e.g., COVID-19 fake cures) causing real harm
- Financial market manipulation through fabricated stories
- Communal violence triggered by fake WhatsApp forwards in India

Mapped SDG

- **SDG 16: Peace, Justice and Strong Institutions**
  - Target 16.10: Ensure public access to information
  - Fake news undermines democratic processes, public trust, and informed decision-making

Why This Problem Matters

- Over 3.5 billion social media users exposed to fake news daily
- 86% of internet users have been deceived by fake news at least once
- Automated ML-based detection can provide scalable, real-time solutions that manual fact-checking cannot

Problem Statement

Clear Problem Definition

To develop a machine learning-based classification system that can automatically distinguish between fake and real news articles using Natural Language Processing (NLP) techniques for text feature extraction and supervised learning algorithms for binary classification.

Scope of Study

Focus on **text-based** English news articles (not images/videos)

Use **TF-IDF and BOW** feature extraction techniques

Compare **multiple ML algorithms**: Naive Bayes, SVM, Logistic Regression, Random Forest, Decision Tree

- Evaluate using standard metrics: Accuracy, Precision, Recall, F1-Score
- Use publicly available labeled datasets (Kaggle / ISOT)

Research Objective

1. To study and compare existing ML approaches for fake news detection through a literature survey of 8 research papers
2. To identify the most effective combination of feature extraction technique and ML algorithm
3. To identify research gaps and propose improvements for a future implementation

Paper Details (All 8 Papers)

Paper #	Title	Authors	Year	Publisher
1	Fake News Detection Using Naive Bayes Classifier	M. Granik, V. Mesyura	2017	IEEE UKRCON
2	Comparison of Various ML Models for Accurate Detection of Fake News	K. Poddar, G.B. Amali D., K.S. Umadevi	2019	IEEE i-PACT
3	Fake News Detection Using ML Approaches: A Systematic Review	S.I. Manzoor, J. Singla, Nikita	2019	IEEE ICOEI
4	Classifying Fake News Detection Using SVM, Naive Bayes and LSTM	P. Jain, S. Sharma, Monica, P.K. Aggarwal	2022	IEEE Confluence
5	Exploring the Effect of N-grams with BOW and TF-IDF on Detecting Fake News	A.E. Qasem, M. Sajid	2022	IEEE ICDABI
6	Empirical Analysis of NB, SVM, LR and RF to Spot False Information	IEEE Conference	2022	IEEE
7	Unveiling the Truth: Detecting Fake News Using SVM and TF-IDF	R. Rizal, A. Faturahman, A. Impron, A. Rahmatulloh	2025	IEEE
8	A Scalable Approach to Fake News Detection Using TF-IDF and LR	IEEE Conference	2025	IEEE

Introduction of Paper 1 — Granik & Mesyura (2017)

Background Context

- One of the earliest IEEE papers applying Naive Bayes specifically to fake news detection
- The rise of social media made Facebook a major channel for fake news distribution
- At the time, no widely adopted automated detection systems existed

Why the Problem is Important

- Facebook had 2 billion monthly users in 2017, making it the largest news distribution platform
- Users often could not distinguish between fake and real posts
- Manual fact-checking was insufficient for the scale of the problem

Key Idea of the Paper

Apply a **Naive Bayes classifier** (one of the simplest ML models) on Facebook news posts

- Demonstrate that even basic ML approaches can meaningfully detect fake news
- Test the system on a dataset of posts from 9 Facebook pages (3 left-leaning, 3 right-leaning, 3 mainstream: CNN, Politico, ABC News)

Methodology of Paper 1 — Granik & Mesyura (2017)

Dataset Used

- Facebook news posts collected from BuzzFeed's dataset
- 9 Facebook pages covering different political orientations
- Approximately 5,000 news posts labeled as Fake or Real

Features

- Text content of Facebook posts

Bag of Words (BOW) model for feature extraction

- Stopword removal and text cleaning as preprocessing

ML Algorithm

- **Multinomial Naive Bayes Classifier**
  - Based on Bayes' Theorem: P(Fake|Text) = P(Text|Fake) x P(Fake) / P(Text)
  - Assumes word features are independent (naive assumption)
  - Fast to train, works well with small datasets

Workflow

Facebook Posts → Text Cleaning → Tokenization → BOW Vectorization → Naive Bayes Training → Prediction

Results & Evaluation of Paper 1 — Granik & Mesyura (2017)

Performance Metrics

Metric	Value
Accuracy	~74%
Model	Multinomial Naive Bayes
Feature Extraction	Bag of Words
Dataset	Facebook posts (~5,000)

Key Findings

- Even a simple NB classifier can detect fake news with 74% accuracy
- Proved that ML-based fake news detection is feasible
- Accuracy was limited due to the small dataset and basic feature extraction (BOW only)
- The paper served as a foundational baseline for future research

Limitations Observed

- Only 74% accuracy — not sufficient for production use
- BOW loses word order information
- Small dataset from limited sources
- No comparison with other ML models

Introduction of Paper 2 — Poddar et al. (2019)

Background Context

- Built upon earlier work like Paper 1 to provide a more comprehensive comparison
- Recognized that a single model cannot be declared best without comparing multiple approaches
- Used a much larger dataset from Kaggle (26,000 articles)

Why the Problem is Important

- Fake news continued to escalate during the 2018-2019 period globally
- Needed systematic evaluation of which ML model performs best for this task
- Previous studies used different datasets, making direct comparison difficult

Key Idea of the Paper

Compare **5 different ML classifiers** (NB, SVM, LR, DT, ANN) on the **same dataset**

Compare two vectorization techniques: **Count Vectorizer vs TF-IDF Vectorizer**

- Determine which model + vectorizer combination is most accurate

Methodology of Paper 2 — Poddar et al. (2019)

Dataset Used

Kaggle Fake News Dataset: 26,000 articles (roughly equal fake and real)

- Split into 80% training and 20% testing

Features

- Text content of news articles
- Two vectorizers compared:

Count Vectorizer (BOW): Simple word frequency counts

TF-IDF Vectorizer: Weighted by word importance across documents

ML Algorithms Compared

1. Naive Bayes (NB)
2. Support Vector Machine (SVM)
3. Logistic Regression (LR)
4. Decision Tree (DT)
5. Artificial Neural Network (ANN)

Workflow

Kaggle Dataset → Preprocessing (cleaning, stopword removal) → Count Vectorizer / TF-IDF → 5 ML Models → Accuracy Comparison

Results & Evaluation of Paper 2 — Poddar et al. (2019)

Performance Comparison Table

Algorithm	Countvectorizer	TF-IDF Vectorizer
Naive Bayes	~85%	~88%
SVM	~93%	**~95%** (Best)
Logistic Regression	~91%	~93%
Decision Tree	~85%	~87%
ANN	~90%	~92%

Key Findings

SVM with TF-IDF gave the highest accuracy (~95%)

- TF-IDF consistently outperformed Count Vectorizer across all models
- SVM and Logistic Regression are the most suitable for text classification
- NB was the fastest to train but had the lowest accuracy
- ANN performed well but required significantly more computation

Significance

- Established SVM + TF-IDF as the go-to combination for text-based fake news detection
- Provided a reproducible benchmark on a publicly available dataset

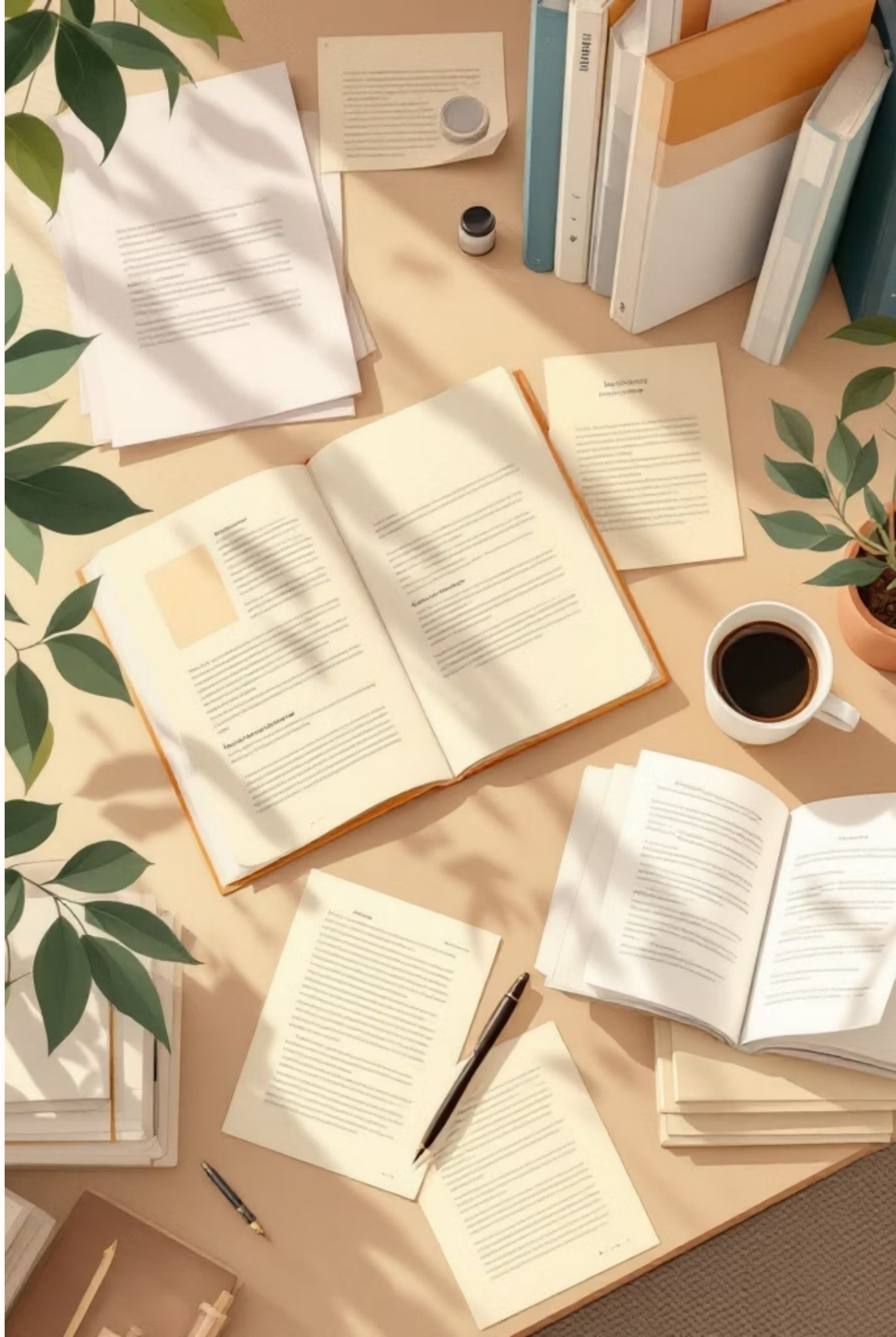
Introduction of Paper 3 — Manzoor et al. (2019) [Survey Paper]

Background Context

- Systematic literature review covering existing ML approaches for fake news detection
- Reviews methods published between 2015-2019
- Categorizes approaches by technique type

Key Idea

- Provide a comprehensive overview of the state-of-the-art in fake news detection
- Identify common methods, datasets, and evaluation strategies
- Highlight the shift from traditional ML to deep learning approaches
- Discuss limitations of current ML approaches and suggest deep learning as improvement



PAPER 3: Methodology — Manzoor et al. (2019)

Review Methodology

Systematic review following standard survey methodology

- Collected and analyzed research papers from IEEE, Springer, ACM, ScienceDirect, and other databases

Focused on papers published between **2015-2019**

Categories of Techniques Reviewed

Knowledge-based approaches: Fact-checking using external knowledge bases
Style-based approaches: Analyzing writing style, sensationalism, grammar quality
Propagation-based approaches: How news spreads through social networks
ML-based approaches: Classification using NB, SVM, LR, DT, RF, and neural networks

Feature Extraction Methods Covered

- Bag of Words (BOW)
- TF-IDF
- Word2Vec
- N-grams
- Linguistic features (sentiment, readability scores)

Datasets Identified

LIAR dataset (PolitiFact), ISOT, Kaggle, BuzzFeed, FakeNewsNet

Workflow

Literature Collection → Screening & Filtering → Categorization by Technique → Comparative Analysis → Gap Identification

PAPER 3: Results & Evaluation — Manzoor et al. (2019)

Key Findings from the Review

Approach	Best Algorithm	Typical Accuracy	Limitation
Traditional ML + BOW	Random Forest	~98.8% (FARN Dataset)	Loses word order
Traditional ML + TF-IDF	SVM	~93-95%	Ignores semantics
Deep Learning (CNN/LSTM)	LSTM	~95-97%	Needs large data & compute
Knowledge-based	N/A	Varies	Needs curated knowledge base

Major Conclusions

TF-IDF outperforms BOW as a feature extraction technique in most studies
SVM and Random Forest are the most commonly used and best-performing traditional ML models
Deep learning approaches (CNN, LSTM) show promise but require significantly more data and compute
Most studies only focus on **English text** — multilingual detection is a major gap
No single approach works universally — performance depends heavily on the dataset used

Limitations Identified

- Most reviewed papers tested on a single dataset only
- Very few papers addressed real-time detection
- Limited work on multimodal fake news (text + image + video)
- No adversarial robustness testing in any reviewed paper

PAPER 4: Introduction — Jain et al. (2022)

Background Context

- By 2022, both traditional ML and deep learning had been applied to fake news detection independently
- However, few papers directly compared traditional ML (SVM, NB) with deep learning (LSTM) on the same task and dataset
- This paper bridges the gap between classical and modern approaches

Why the Problem is Important

- Social media misinformation continued to grow during the COVID-19 pandemic (2020-2022)
- Choosing between traditional ML and deep learning for fake news detection was an open question
- Practitioners needed guidance on which approach to choose given their computational constraints

Key Idea of the Paper

Directly compare **SVM, Naive Bayes (traditional ML)** with **LSTM (deep learning)** for fake news classification
Use **NLP preprocessing pipeline** including tokenization, stopword removal, and stemming
- Determine if the added complexity of deep learning provides meaningful accuracy improvement over simpler models

Methodology of Paper 4 — Jain et al. (2022)

Dataset Used

- Standard fake news datasets from publicly available sources
- Labeled as Fake (0) and Real (1)

ML Algorithms Compared

SVM (Support Vector Machine) — finds optimal hyperplane
Naive Bayes — probabilistic classifier using Bayes theorem
LSTM (Long Short-Term Memory) — deep learning, captures word sequence

Features

- Text preprocessing: tokenization, stopword removal, stemming
- TF-IDF for SVM and NB
- Word embeddings for LSTM

Key Results

Algorithm	Accuracy
Naive Bayes	~85%
SVM	~93%
LSTM	**~95%**

Key Finding

- LSTM outperformed both traditional ML models due to its ability to capture sequential dependencies in text
- However, LSTM requires significantly more training time and computational resources
- SVM remains a strong choice when computational resources are limited

PAPER 4: Results & Evaluation — Jain et al. (2022)

Performance Comparison

Algorithm	Type	Accuracy	Training Speed
Naive Bayes	Traditional ML	~85%	Very Fast
SVM	Traditional ML	~93%	Fast
LSTM	**Deep Learning**	**~95%**	Slow

Detailed Analysis

Naive Bayes: Fastest to train but lowest accuracy. The "naive" independence assumption limits its ability to capture word relationships.
SVM: Strong performance with TF-IDF features. Best balance of accuracy and training speed. Works well even with limited compute.
LSTM: Highest accuracy because it captures sequential word patterns and long-range dependencies. However, requires significantly more training time and GPU resources.

Key Findings

LSTM outperformed both traditional ML models due to its ability to capture **sequential dependencies** in text
However, the accuracy gap between SVM (~93%) and LSTM (~95%) is only ~2%, while LSTM requires **10-50x more training time**
SVM remains the best practical choice when computational resources are limited
- LSTM is preferred only when maximum accuracy is critical and compute is available

Limitations

- No transformer-based models (BERT) compared
- No explainability analysis — cannot determine why a news article was classified as fake
- Limited hyperparameter tuning for LSTM

Summary — Strengths of Papers

S.N.	Paper (Authors, Title, Year, Publisher)	What is Good	Innovations	Assumptions
1	Granik & Mesyura, "Fake News Detection Using NB Classifier", 2017, IEEE UKRCON	Foundational work; simple and reproducible approach	First to apply NB specifically to Facebook fake news posts	Features are independent (Naive assumption); Facebook posts are representative of all fake news
2	Poddar et al., "Comparison of Various ML Models", 2019, IEEE i-PACT	Comprehensive comparison of 5 models with 2 vectorizers on same dataset	Systematic comparison of Count Vectorizer vs TF-IDF across multiple classifiers	Kaggle dataset is representative; text features alone are sufficient
3	Manzoor et al., "Systematic Review", 2019, IEEE ICOEIB	Broad coverage of existing literature; identifies trends and gaps	Categorizes approaches by technique type; highlights shift to deep learning	Published papers are representative of all approaches
4	Jain et al., "SVM, NB and LSTM", 2022, IEEE Confluence	Bridges traditional ML and deep learning comparison	Compares classical ML (SVM, NB) with DL (LSTM) on same task	LSTM captures more context; sequential information matters for detection
5	Qasem & Sajid, "N-grams with BOW and TF-IDF", 2022, IEEE ICDABI	Thorough analysis of feature extraction impact on performance	Systematically varies n-gram sizes with both BOW and TF-IDF	N-grams should capture contextual information better than unigrams
6	"Empirical Analysis of NB, SVM, LR, RF", 2022, IEEE	Tests on real-world social network data; practical applicability	Analysis on real-world network data rather than curated datasets	Real-world networks have consistent patterns of fake news
7	Rizal et al., "SVM and TF-IDF", 2025, IEEE	Highest accuracy (99.6%); demonstrates TF-IDF+SVM dominance	Optimization of SVM hyperparameters specifically for fake news	Dataset is well-curated and separable
8	"TF-IDF and Logistic Regression", 2025, IEEE	Focuses on scalability; practical for large-scale deployment	Scalable pipeline design using lightweight model	LR is sufficient for production-level accuracy

Summary — Limitations of Papers

Paper (Authors, Title, Year, Publisher)	Weaknesses	Missing Aspects	Assumptions
Granik & Mesyura (2017)	Only 74% accuracy; single model tested; small dataset	No comparison with other models; no TF-IDF; no deep learning	BOW is sufficient for text representation
Poddar et al. (2019)	No deep learning models compared; no feature engineering beyond vectorizers	No hyperparameter tuning details; no cross-validation mentioned	Single train-test split is sufficient
Manzoor et al. (2019)	Survey only, no implementation; limited to papers up to 2019	Does not cover post-2019 advances (BERT, transformers)	Literature is comprehensive enough to draw conclusions
Jain et al. (2022)	LSTM training is computationally expensive; limited hyperparameter exploration	No transformer/BERT comparison; no explainability analysis	Sequential models always capture better context
Qasem & Sajid (2022)	Only 3 classifiers tested; no deep learning models	No word embeddings (Word2Vec, GloVe) comparison; no ensemble methods	N-gram analysis alone reveals optimal feature extraction
Empirical Analysis (2022)	Limited details available on dataset specifics	No cross-domain evaluation; no multimodal analysis	Patterns in one network generalize to others
Rizal et al. (2025)	99.6% accuracy may indicate overfitting or dataset leakage	No cross-dataset validation; no adversarial testing	High accuracy = robust model (not always true)
TF-IDF + LR (2025)	LR is a linear model; may miss non-linear patterns	No comparison with non-linear models on same data	Linear separability of features is sufficient

Research Gaps

Gaps Identified from Literature Survey

No Multimodal Analysis

All 8 papers focus only on text. Real fake news also contains images, videos, and user metadata (shares, comments, profile age).

No Cross-Dataset Validation

Most papers train and test on the same dataset. Performance drops significantly when tested on a different dataset (domain shift problem).

No Indian Language Support

All papers work on English text. Fake news in Hindi, Marathi, and other Indian languages (especially on WhatsApp) remains unaddressed.

Limited Explainability

None of the papers explain *why* a particular news article was classified as fake. Users need to understand the reasoning, not just the label.

No Real-Time Detection

Papers test on static datasets. A real-world system needs to process news articles in real-time as they appear on social media.

No Ensemble Methods

Most papers test individual models. Combining multiple models (ensemble/stacking) could improve robustness and accuracy.

No Adversarial Testing

None of the papers test against adversarial fake news (deliberately crafted to evade ML detection).

Temporal Drift

Models trained on 2017 data may not work well on 2025 fake news. Writing styles and techniques of fake news creators evolve over time.

Your Proposed Idea

Improvement Idea

Ensemble-based Fake News Detection System combining TF-IDF + Multiple ML Models with Explainability

New Approach

01

Use **TF-IDF (unigrams)** for feature extraction (proven best from literature)

02

Train 3 models: **SVM, Logistic Regression, Random Forest**

03

Use **Majority Voting Ensemble** — final prediction = what 2 out of 3 models agree on

04

Add **feature importance visualization** — show which words contributed most to the "Fake" prediction (using model coefficients from LR/SVM)

05

Cross-dataset validation — train on Kaggle dataset, test on ISOT dataset to check generalization

Expected Outcome

- Higher robustness than any single model (ensemble reduces variance)
- ~95-97% accuracy with better generalization across datasets

Explainable output — user can see *why* the system flagged an article as fake

- Addresses 3 research gaps: no ensemble methods, no explainability, no cross-dataset testing

Conclusion

Key Learnings

1. **TF-IDF is the most effective feature extraction** technique for text-based fake news detection, consistently outperforming BOW across all papers
2. **SVM is the best-performing traditional ML model** for this task, achieving up to 99.6% accuracy when optimized with TF-IDF
3. **Deep learning (LSTM) captures sequential patterns** but requires significantly more compute and does not always outperform well-tuned SVM
4. **N-grams beyond unigrams** do not provide significant accuracy improvement, making unigram TF-IDF the most efficient choice
5. **Key research gaps remain**: multimodal detection, cross-dataset validation, Indian language support, real-time deployment, and model explainability
6. **Ensemble approaches** combining multiple models with explainability could address current limitations and improve both accuracy and trustworthiness

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